MINDERA EDUCATION | CODE ACADEMY

# Test Automation

## Backend

1st edition

## Module 2: Request Types

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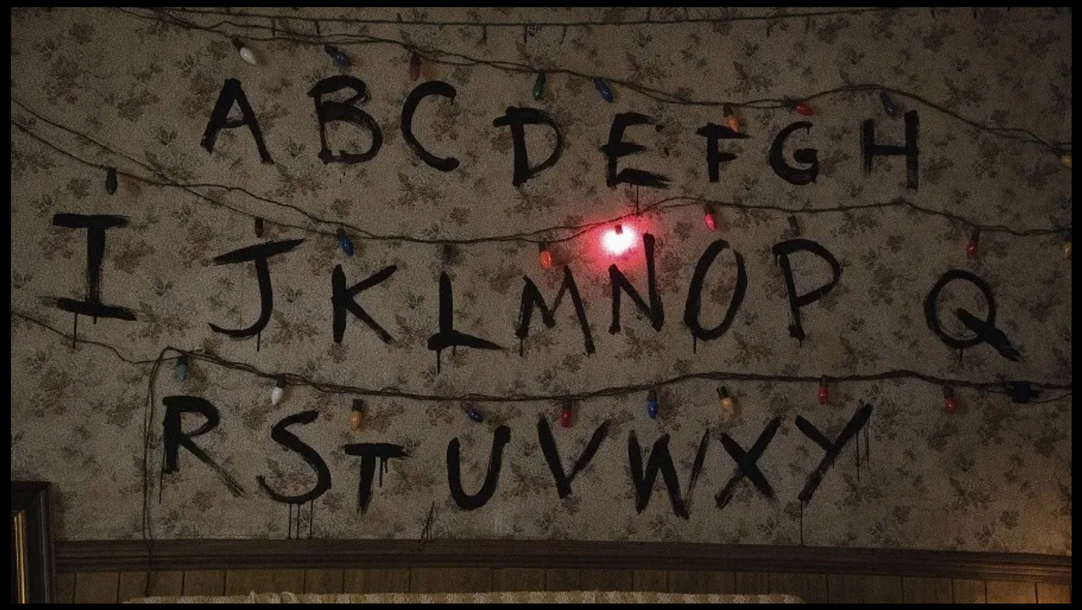
# HTTP Methods

HTTP Methods help communicate which **action** you want to perform with the data on the web.

| | |
|---|---|
| GET | POST |
| PUT | DELETE |
| PATCH | HEAD |
| OPTIONS | TRACE |
| CONNECT | |

# GET

GET is used to **retrieve or view data** that is already available, such as viewing a product page, a list of movies, or user details, without modifying or adding anything.



# GET

ENDPOINT

```
GET /api/characters/will-byers
```

RESPONSE


```
{
  "name": "Will Byers",
  "status": "Missing",
  "location": "Upside Down",
  "notes": "Trapped in the Upside Down, unable to communicate"
}
```

# POST

**POST** is used to **send data** to the server, creating or updating a resource. For example, it can be used to add a new product, register a user, or send a message, modifying or adding information on the server.

# POST

## ENDPOINT

POST /api/characters/hopper/teleport

## REQUEST BODY

```
{
  "name": "Jim Hopper",
  "role": "Sheriff",
  "location": "Hawkins",
  "skills": ["combat training", "investigation", "leadership"],
  "status": "active"
}
```

## RESPONSE BODY

```
{
  "message": "Jim Hopper successfully sent to the Upside Down",
  "status": "success",
  "character": {
    "name": "Jim Hopper",
    "role": "Sheriff",
    "status": "active",
    "destination": "Upside Down",
    "mission": "Rescue"
  }
}
```

# PUT

PUT is used to **update** data on the server, completely replacing an existing resource or creating a new one if it doesn’t already exist.

# PUT

## ENDPOINT

PUT /api/characters/will-byers

## REQUEST BODY


```
{
  "name": "Will Byers",
  "status": "Recovered",
  "location": "Hawkins",
  "notes": "Experiencing Upside Down visions, but physically safe"
}
```


## RESPONSE BODY


```
{
  "message": "Will Byers' status successfully updated",
  "status": "success",
  "character": {
    "name": "Will Byers",
    "status": "Recovered",
    "location": "Hawkins",
    "notes": "Experiencing Upside Down visions, but physically safe"
  }
}
```

# DELETE

**DELETE** is used to **remove** a resource from the server.

When using **DELETE**, you are telling the server that a specific resource (such as a character, an item, or an entity) should be **permanently deleted**.

# DELETE

## ENDPOINT

```
DELETE /api/creatures/demogorgon
```

## RESPONSE BODY


```
{
  "message": "Demogorgon has been successfully deleted",
  "status": "success",
  "creature": {
    "name": "Demogorgon",
    "danger_level": "extreme",
    "origin_location": "Upside Down",
    "status": "removed"
  }
}
```

# PATCH

PATCH is used to partially update existing data, meaning you can modify only specific fields of a resource without replacing the entire object.

It is commonly used when you want to update just a few details, such as changing a user’s email or updating a character’s level, without affecting the rest of the data.

# PATCH

## ENDPOINT

PATCH /api/characters/will-byers

## REQUEST

```
{
  "status": "Found"
}
```

## RESPONSE

```
{
  "message": "Character updated successfully",
  "status": "success",
  "character": {
    "name": "Will Byers",
    "status": "Found",
    "location": "Hawkins",
    "notes": "Recovered from the Upside Down"
  }
}
```

# STATUS CODE

Status codes are numbers returned by the server to indicate the result of a request. They help communicate whether the operation was successful or if there was a problem.

| | | | |
|---|---|---|---|
| **1xx** | INFORMATIVE | **Request received and being processed.** | **100 Continue** |
| 2xx | SUCCESS | Request sucessfull | 200 OK, 201 Created |
| 3xx | REDIRECT | Request needs more actions to be continued. | 301 Moved Permanently |
| 4xx | CLIENT ERROR | Invalid request made by the client. | 400 Bad Request, 404 Not Found |
| 5xx | SERVER ERROR | Server failed to process this request. | 500 Internal Server Error |